Kurzgesagt: What is an Atom? (Educational Science Lesson) | Student Workbook

Student Name: ________________________ **Date:** ______________ **Class:** ____________

Summary

Atoms are the basic building blocks of all matter in the universe, consisting of protons, neutrons, and electrons. The center of an atom is the nucleus, which holds the positive protons and neutral neutrons. Electrons orbit the nucleus in shells, creating chemical bonds with other atoms to form molecules.

Questions

Q1. According to the video, what are the three basic subatomic particles that make up an atom?

Q2. Where in the atom is the nucleus located, and which particles does it contain?

Q3. What is the electric charge of an electron, and where are they found within the atom?

Q4. Why do electrons orbit the nucleus rather than fly off into space?

Q5. What happens when atoms share or exchange electrons with neighboring atoms?

Kurzgesagt: What is an Atom? (Educational Science Lesson) | Teacher Answer Key

Summary:

Atoms are the basic building blocks of all matter in the universe, consisting of protons, neutrons, and electrons. The center of an atom is the nucleus, which holds the positive protons and neutral neutrons. Electrons orbit the nucleus in shells, creating chemical bonds with other atoms to form molecules.

Questions:

Q1. According to the video, what are the three basic subatomic particles that make up an atom?

Answer: The three subatomic particles are protons (positively charged), neutrons (neutrally charged), and electrons (negatively charged).

Q2. Where in the atom is the nucleus located, and which particles does it contain?

Answer: The nucleus is located at the center of the atom. It contains protons and neutrons clumped tightly together.

Q3. What is the electric charge of an electron, and where are they found within the atom?

Answer: Electrons have a negative electric charge. They are found orbiting the nucleus in complex electron shells/clouds.

Q4. Why do electrons orbit the nucleus rather than fly off into space?

Answer: Electrons are attracted to the positive charge of the protons in the nucleus (electromagnetic force), which keeps them bound in orbit.

Q5. What happens when atoms share or exchange electrons with neighboring atoms?

Answer: Sharing or exchanging electrons creates chemical bonds, allowing atoms to combine and form molecules like water or carbon dioxide.

The Mighty Atom Team Trivia!

Host Instructions for Teachers:

Split the classroom into groups of 3-4. Read each trivia question aloud. Teams have 30 seconds to write down their answer.

ROUND 1: Which of the following subatomic particles has a positive charge?

Options: • A) Electron • B) Neutron • C) Proton • D) Photon

Correct Option: [C (Proton - Protons are positive, electrons are negative, neutrons have no charge).]

ROUND 2: What is the center core of an atom called?

Options: • A) Cell • B) Nucleus • C) Shell • D) Orbit

Correct Option: [B (Nucleus - The nucleus contains the mass of the atom, holding protons and neutrons).]

ROUND 3: True or False: Electrons are much larger and heavier than protons.

Options: • A) True • B) False

Correct Option: [B (False - Electrons are extremely tiny compared to protons and neutrons; they are roughly 1800 times lighter).]

ROUND 4: What holds two or more atoms together in a molecule?

Options: • A) Gravity • B) Chemical bonds • C) Nuclear fission • D) Magnetic glue

Correct Option: [B (Chemical bonds - Atoms share or exchange valence electrons to create tight chemical structures).]

ROUND 5: Which Greek word does 'Atom' originate from, meaning indivisible?

Options: • A) Atomos • B) Arche • C) Sophia • D) Bios

Correct Option: [A (Atomos - Ancient Greek philosopher Democritus coined the term 'atomos' to describe the smallest uncuttable unit of matter).]